

Multiply up to four digits by two digits

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Calculate $2,304 \times 27$. Copy or complete the first step.

2. Calculate $4,216 \times 34$.

3. Calculate $3,075 \times 46$.

4. Miro says: Miro treats the 2 in 27 as 2 rather than 20. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate $2,304 \times 27$. Copy or complete the first step.
Answer 62,208.
2. Calculate $4,216 \times 34$.
Answer 143,344.
3. Calculate $3,075 \times 46$.
Answer 141,450.
4. Miro says: Miro treats the 2 in 27 as 2 rather than 20. Is this correct? Explain.
Answer The 2 represents two tens, so its partial product is twenty times the first factor.

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Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Calculate $2,304 \times 27$.

6. Check this result using an inverse, estimate or known fact: Calculate $3,075 \times 46$.

2. Calculate $4,216 \times 34$.

7. Use compensation to calculate $4,999 \times 62$.

3. Calculate $3,075 \times 46$.

8. Identify and correct this misconception: Miro treats the 2 in 27 as 2 rather than 20.

4. Solve this independently and show every stage: Calculate $2,304 \times 27$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate $4,216 \times 34$.

10. Write and solve a similar question to: Calculate $3,075 \times 46$.

Teacher answers and checking prompts

1. Calculate $2,304 \times 27$.
Answer 62,208.
2. Calculate $4,216 \times 34$.
Answer 143,344.
3. Calculate $3,075 \times 46$.
Answer 141,450.
4. Solve this independently and show every stage: Calculate $2,304 \times 27$.
Answer 62,208.
5. Use a second method or representation for: Calculate $4,216 \times 34$.
Answer Expected result: 143,344. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Calculate $3,075 \times 46$.
Answer Expected result: 141,450. A valid check must be shown.
7. Use compensation to calculate $4,999 \times 62$.
Answer 309,938.
8. Identify and correct this misconception: Miro treats the 2 in 27 as 2 rather than 20.
Answer The 2 represents two tens, so its partial product is twenty times the first factor.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Calculate $3,075 \times 46$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Multiply up to four digits by two digits

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Use compensation to calculate $4,999 \times 62$.

6. Create a new example that tests the same idea as: Calculate $3,075 \times 46$.

2. Prove the answer to this question, rather than only calculating it: Calculate $2,304 \times 27$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate $4,216 \times 34$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 141,450.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro treats the 2 in 27 as 2 rather than 20.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Use compensation to calculate $4,999 \times 62$.
Answer 309,938.
2. Prove the answer to this question, rather than only calculating it: Calculate $2,304 \times 27$.
Answer Expected result: 62,208. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate $4,216 \times 34$.
Answer Expected result: 143,344. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 141,450.
Answer One valid reconstruction is: Calculate $3,075 \times 46$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro treats the 2 in 27 as 2 rather than 20.
Answer The 2 represents two tens, so its partial product is twenty times the first factor.
6. Create a new example that tests the same idea as: Calculate $3,075 \times 46$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.